Proficient in ETABS (80 Hours)

S.No	Session	Topics Covered	Type	Duration (hrs)
1	Session 1	Overview of Structural Analysis and Design Introduction Workflow of ETABS ETABS User Interface	Theory/Demo	2.0 Hrs
2	Session 2	Excercise included for practice	Practical	2.0 Hrs
3	Session 3	**Creating New Project:** New Model Quick Template, Edit Stories and Grid System, Add Grid at Selected Joints,Grid Options **Material Properties:** Defining Materials, Material Property Data **Section Properties:** Defining Various Sections, Section Property Data	Theory/Demo	2.0 Hrs
4	Session 4	Excercise included for practice	Practical	2.0 Hrs
5	Session 5	**Modeling:** Draw Joint Objects, Draw Beam/Column/Brace Objects, Draw Floor/Wall Objects **Modeling:** Draw Reference Points, Draw Reference Planes, Draw Wall Stacks	Theory/Demo	2.0 Hrs
6	Session 6	Excercise included for practice	Practical	2.0 Hrs
7	Session 7	**Object Selection Tools:** Select Tool, Deselect Tool **Replicate:** Linear, Radial, Mirror, Story **Extrude:** Joint to Frame, Frame to Shell Merge Joints Align Joints/Frames/Edges Move Joints /Frames/Edges	Theory/Demo	2.0 Hrs
8	Session 8	Excercise included for practice	Practical	2.0 Hrs
9	Session 9	**Edit Frames:** Divide Frames, Merge Frames, Reverse Frame Connectivity, Modify/Show Frame Type **Edit Shells:** Divide Shells, Merge Shells, Expand/Shrink Shells, Split Shell Edge, Remove Joints from Shells,Chamfer Slab Corners, Reverse Wall Local 3 Axis, Modify/show Wall Curve Type	Theory/Demo	2.0 Hrs
10	Session 10	Excercise included for practice	Practical	2.0 Hrs
11	Session 11	Assigning Section Properties Support Specification	Theory/Demo	2.0 Hrs
12	Session 12	Excercise included for practice	Practical	2.0 Hrs
13	Session 13	Loading, Analysis & Result Define Load Pattern, Shell Uniform Load Sets, Joint Loads, Frame Loads, Shell Loads Assign Loads Load Cases, Load Combinations	Theory/Demo	2.0 Hrs
14	Session 14	Excercise included for practice	Practical	2.0 Hrs
15	Session 15	Manual Calculations (Dead Load & Live Load) Set Load Cases to Run, Check Model, Analysis **Display Result:** Deformed Shapes, BMD, SFD, Shell Stresses, Tabular Reports, Support Spring Reactions **Display:** Undeformed Shape, Joint Load Assigns, Frame Load Assigns, Shell Load Assigns	Theory/Demo	2.0 Hrs
16	Session 16	Excercise included for practice	Practical	2.0 Hrs
17	Session 17	**Concrete Frame Design:** Introduction, Design Procedure, View/Revise Preferences, View/Revise Overwrites,Select Design Groups, Select Design Combinations, Start Design Check, Interactive Design, Display Design Info,Change Design Section, Verify Analysis Vs Design Section, Verify All Members Passed	Theory/Demo	2.0 Hrs
18	Session 18	Excercise included for practice	Practical	2.0 Hrs
19	Session 19	Group Definitions Shear Wall Design	Theory/Demo	2.0 Hrs
20	Session 20	Excercise included for practice	Practical	2.0 Hrs
21	Session 21	**Steel Frame Design:** Introduction, Design Procedure, View/Revise Preferences, View/Revise Overwrites, SelectDesign Combinations **Braces:** Lateral, Uniform, Point Set Lateral Displacement Targets Start Design/Check Interactive Design	Theory/Demo	2.0 Hrs
22	Session 22	Excercise included for practice	Practical	2.0 Hrs
		Steel Connection Design: Introduction, Design Procedure		

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23	Session 23	**Steel Joist Design:** Introduction, Design Procedure, View/Revise Preference, View/Revise Overwrites, SelectDesign Groups, Select Design Combinations, Start Design/Check, Interactive Design	Theory/Demo	2.0 Hrs
24	Session 24	Excercise included for practice	Practical	2.0 Hrs
25	Session 25	**Flat Slab Design:** Introduction Components of Slabs, Structural Geometry and Properties for Flat Slab, Structural System Properties, Restraints at Bottom, Loads, Floor Diaphragm Rigidity	Theory/Demo	2.0 Hrs
26	Session 26	Excercise included for practice	Practical	2.0 Hrs
27	Session 27	**Waffle Slab Design:** Introduction Structural Geometry and Properties for Flat Slab Structural System Properties, Restraints at Bottom, Loads, Floor Diaphragm Rigidity	Theory/Demo	2.0 Hrs
28	Session 28	Excercise included for practice	Practical	2.0 Hrs
29	Session 29	Static Linear Analysis Introduction, Manual Calculation Define Load Patterns, Modify Lateral Load Mass Source, Mass Source Data, Modal Cases, Modal Case Data	Theory/Demo	2.0 Hrs
30	Session 30	Excercise included for practice	Practical	2.0 Hrs
31	Session 31	Dynamic Linear Analysis Introduction, Methodology, Terminology Response Spectrum Function, Story Response-Maximum Story Displacement	Theory/Demo	2.0 Hrs
32	Session 32	Excercise included for practice	Practical	2.0 Hrs
33	Session 33	Non-Linear Analysis Introduction to Linear Vs Non-Linear **Push Over Analysis:** Introduction, General Procedure	Theory/Demo	2.0 Hrs
34	Session 34	Excercise included for practice	Practical	2.0 Hrs
35	Session 35	**Time History Analysis:** Introduction Time History Function, Output P-Delta Effect, Practical Application	Theory/Demo	2.0 Hrs
36	Session 36	Excercise included for practice	Practical	2.0 Hrs
37	Session 37	Detailing Detailing Preferences, Concrete Component Preferences, Steel Component Preferences Rebar Selection Rules, Start Detailing, Show Detailing, Clear Detailing **Custom Section Designer:** Introduction, Section Designer Workspace	Theory/Demo	2.0 Hrs
38	Session 38	Excercise included for practice	Practical	2.0 Hrs
39	Session 39	Export Report Creation Print Graphics, Show Input / Output Text Files	Theory/Demo	2.0 Hrs
40	Session 40	Excercise included for practice	Practical	2.0 Hrs